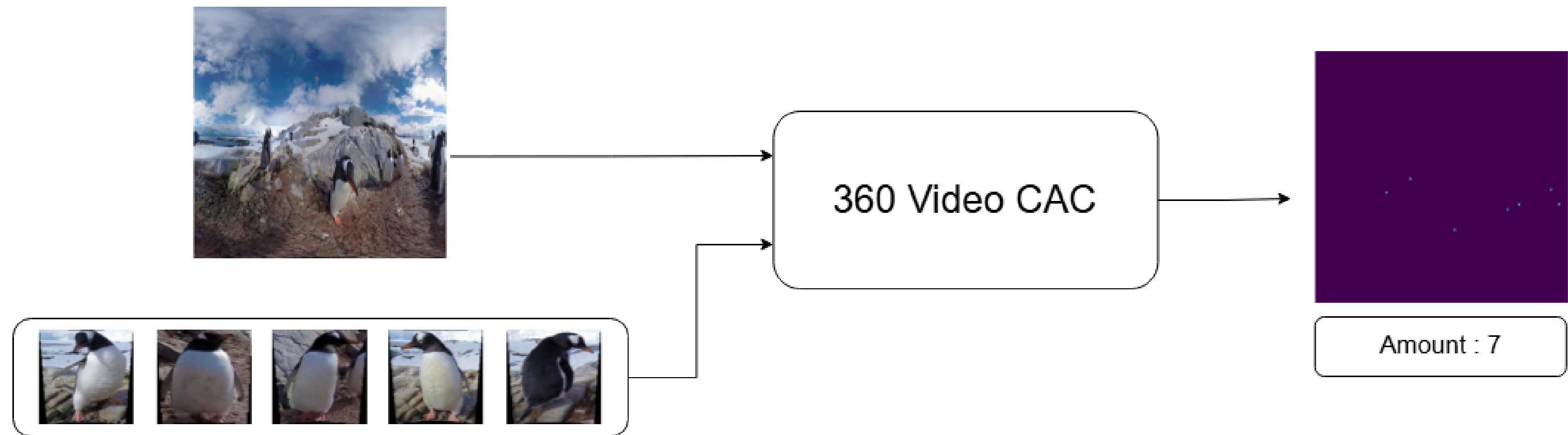


專案 – 360 Video CAC



專案背景



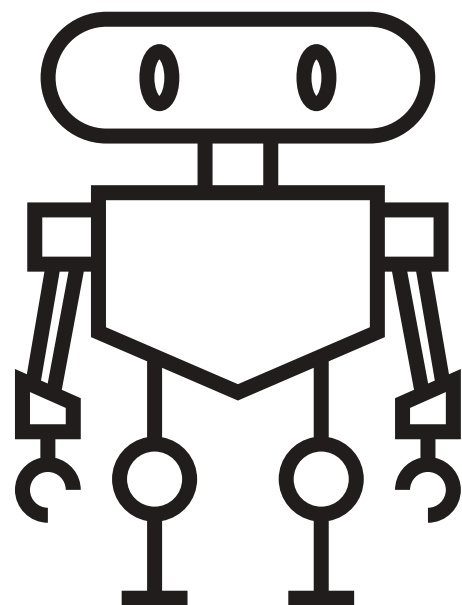
360 Video



ERP Image

專案應用

Robotics



VR 領域



專案動機

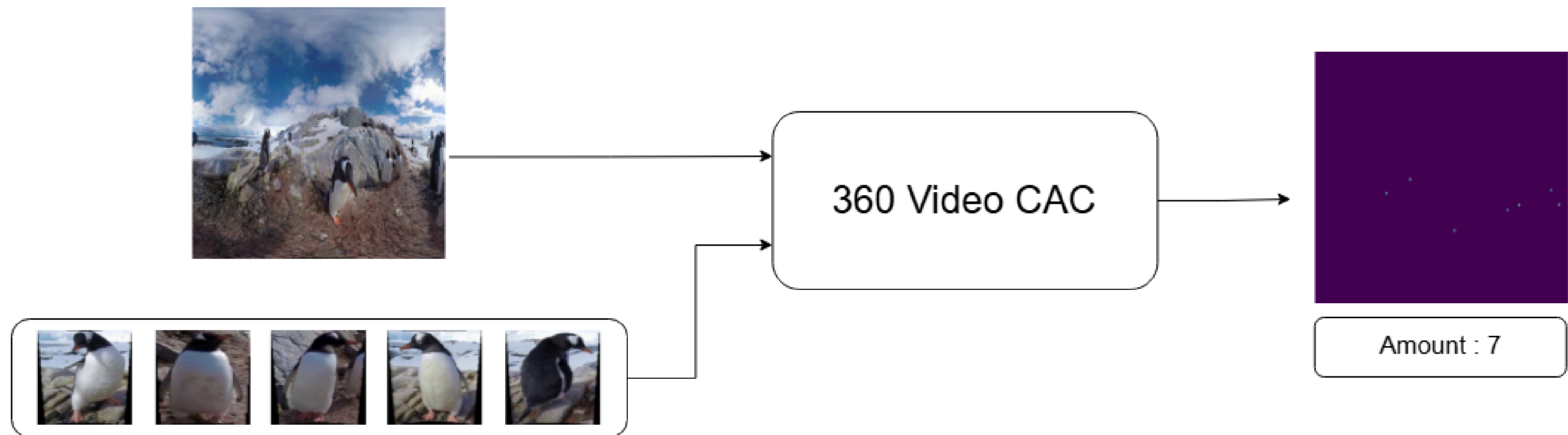
節省投影成本

現存的方法需要先將 ERP Image 投影到平面，耗費投影時間

增加模型泛用性

現存的方法只能計算某些特定種類的物體數量，泛用性不足

專案目的



實驗設計 – 確保泛用性

Training Set

- 電腦
- 汽車
- 椅子
- 窗戶
- 書本
- 櫃子

Validating Set

- 門
- 企鵝
- 瓶子

Testing Set

- 狗
- 船
- 人

實驗設計 – 兩階段實驗

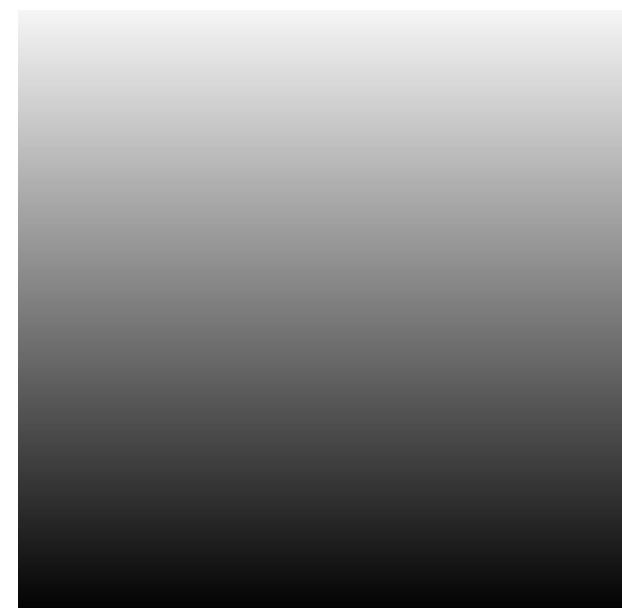
Stage 1

修改 CFOCNet

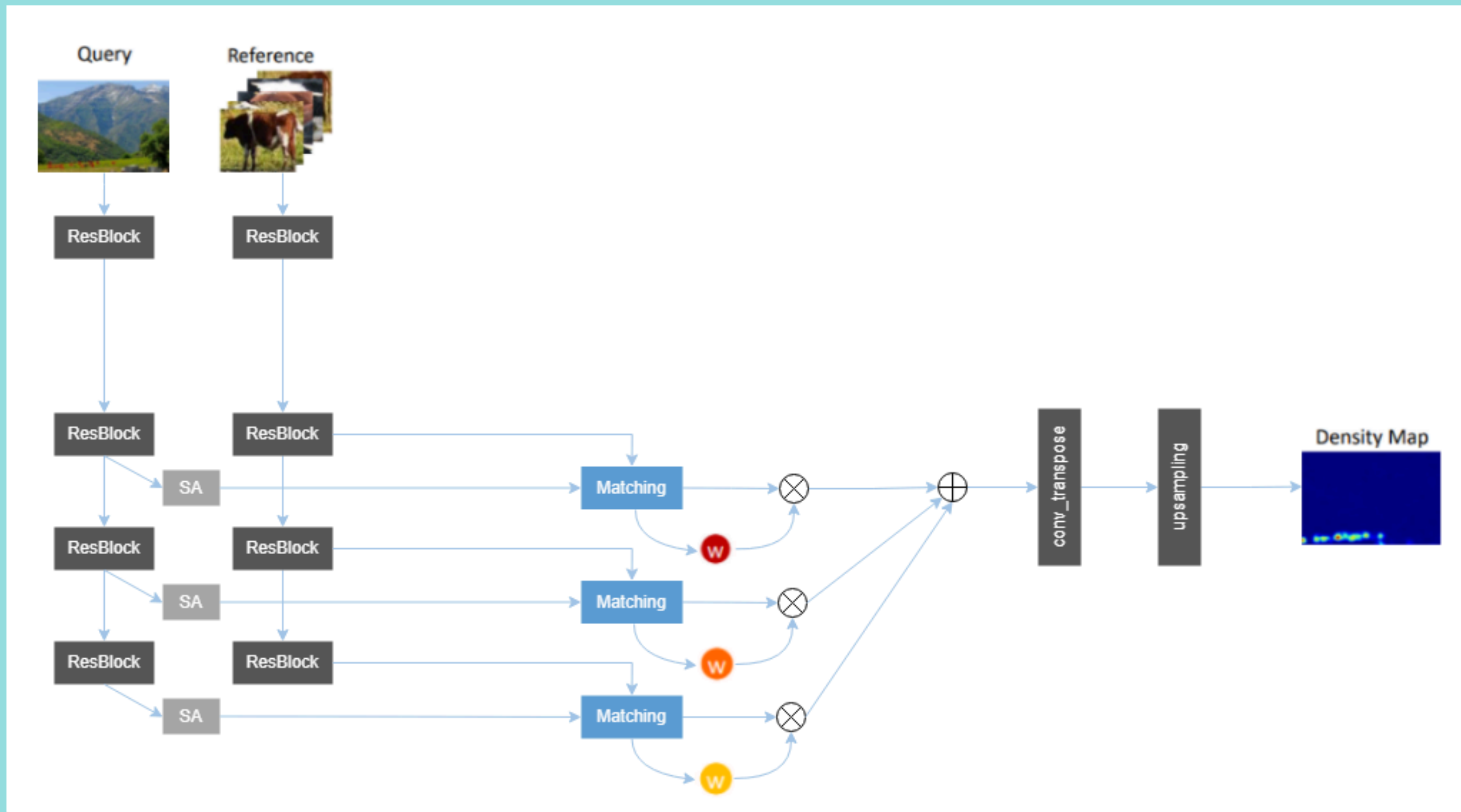
1. L2 Normalization
2. DropBlock
3. Huber Loss
4. Focal Loss
5. Granularity Fusion

Stage 2

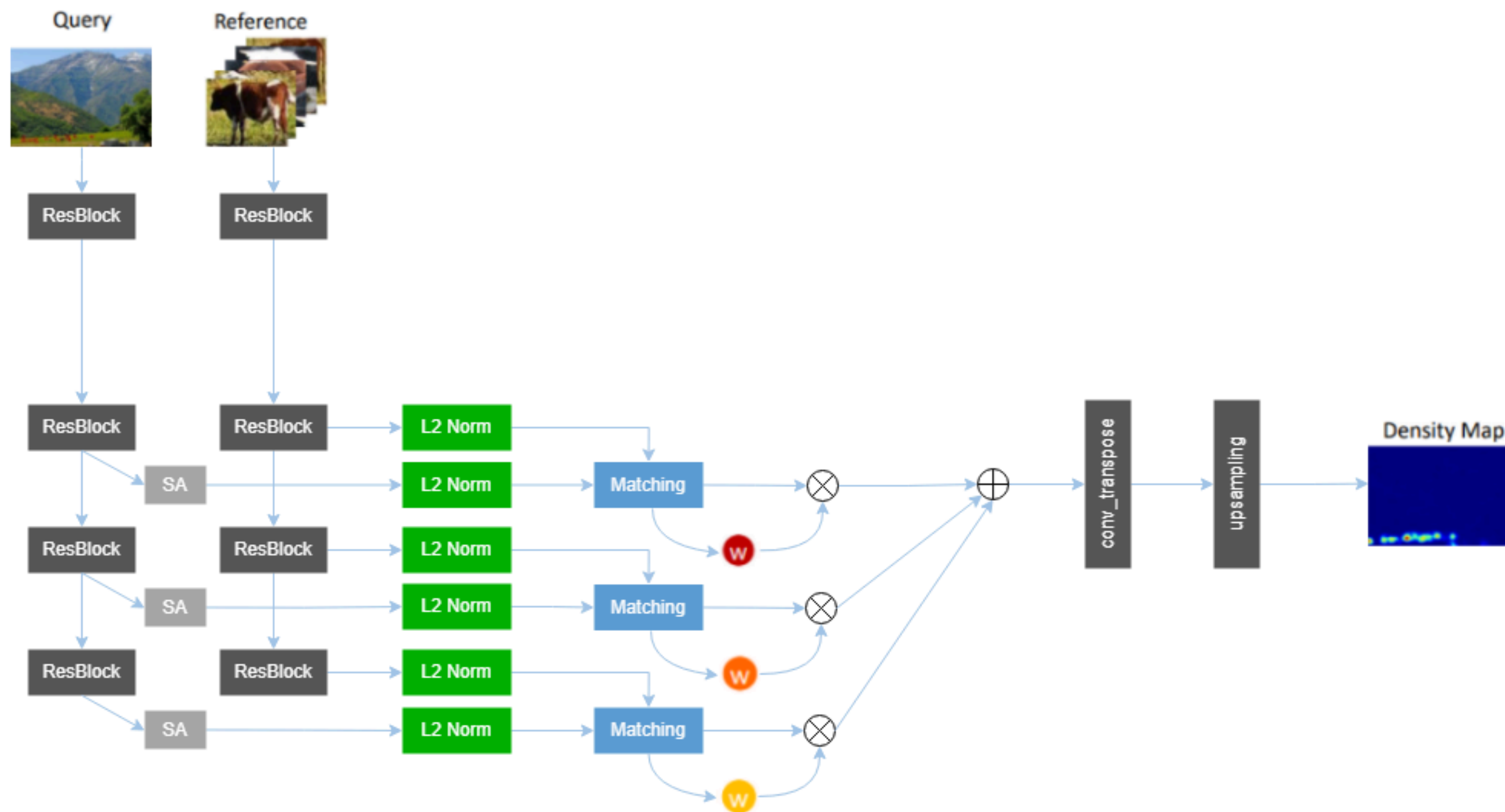
增加變形 Feature



Base Model – CFOCNet



改良方法 – L2 Norm



改良方法 – L2 Norm

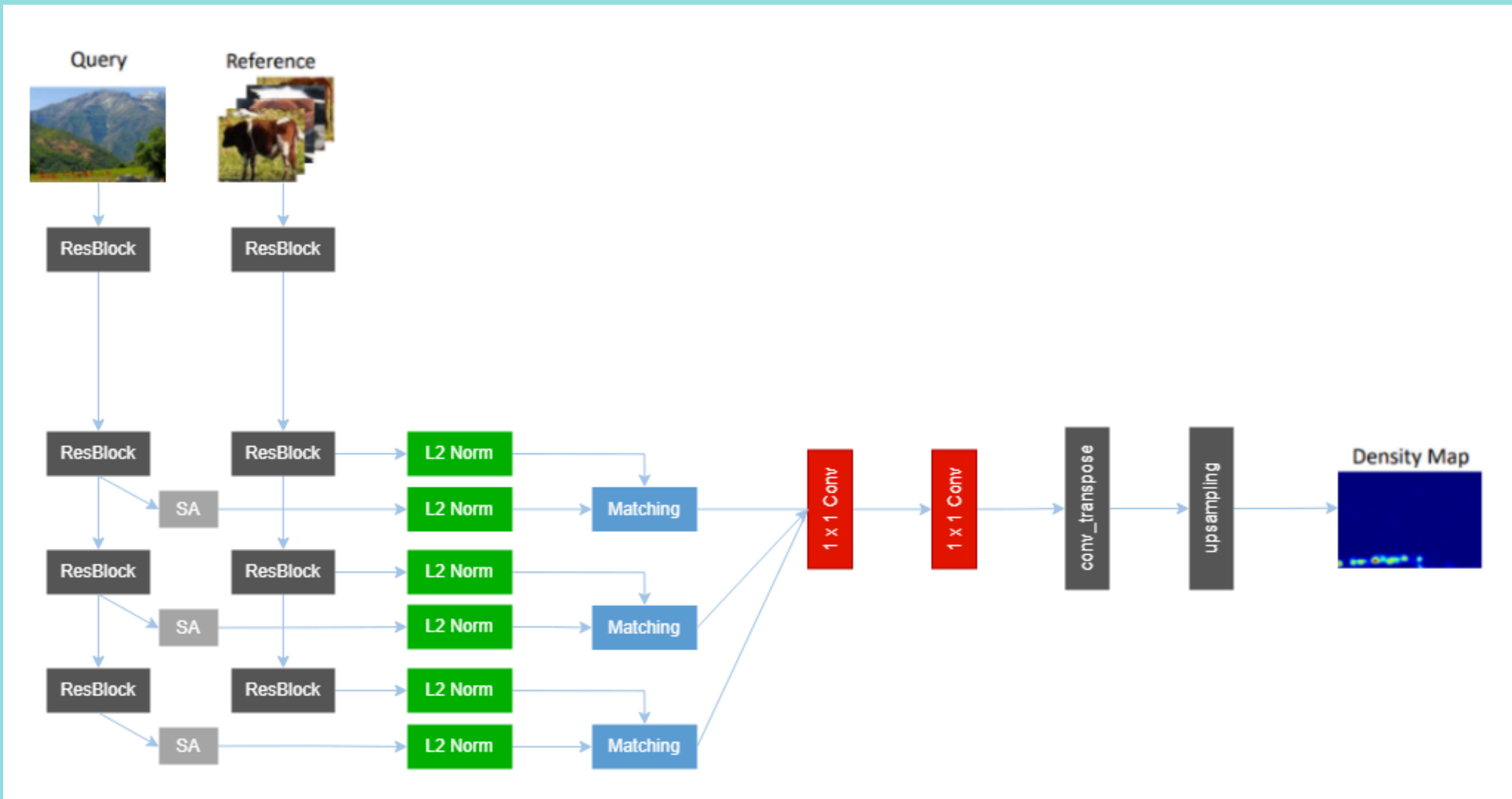
具體方法

$$v = \frac{v}{\max(\|v\|_p, \epsilon)}$$

改良結果

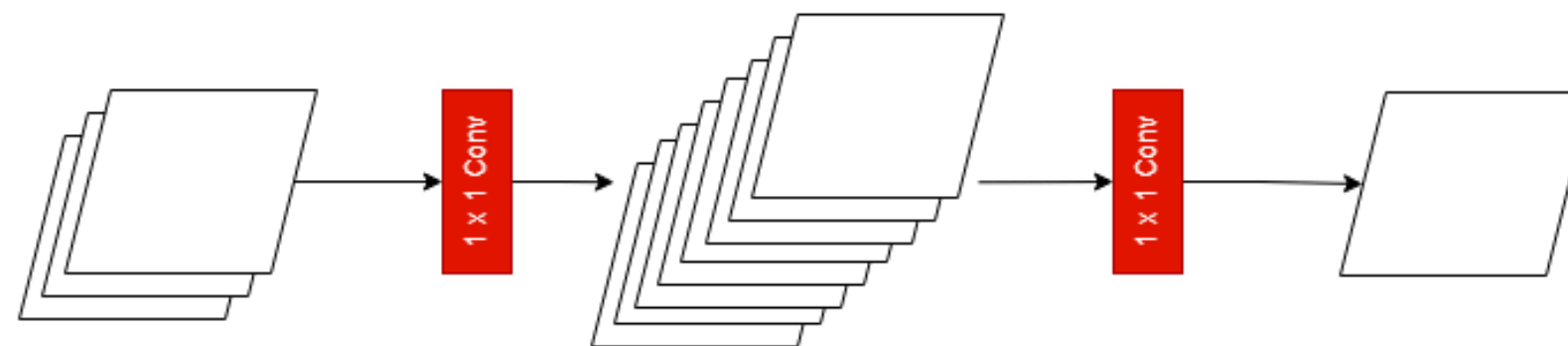
只需要原本一半的 step 就可以完成訓練

改良方法 – Pointwise Convolution



改良方法 – Pointwise Convolution

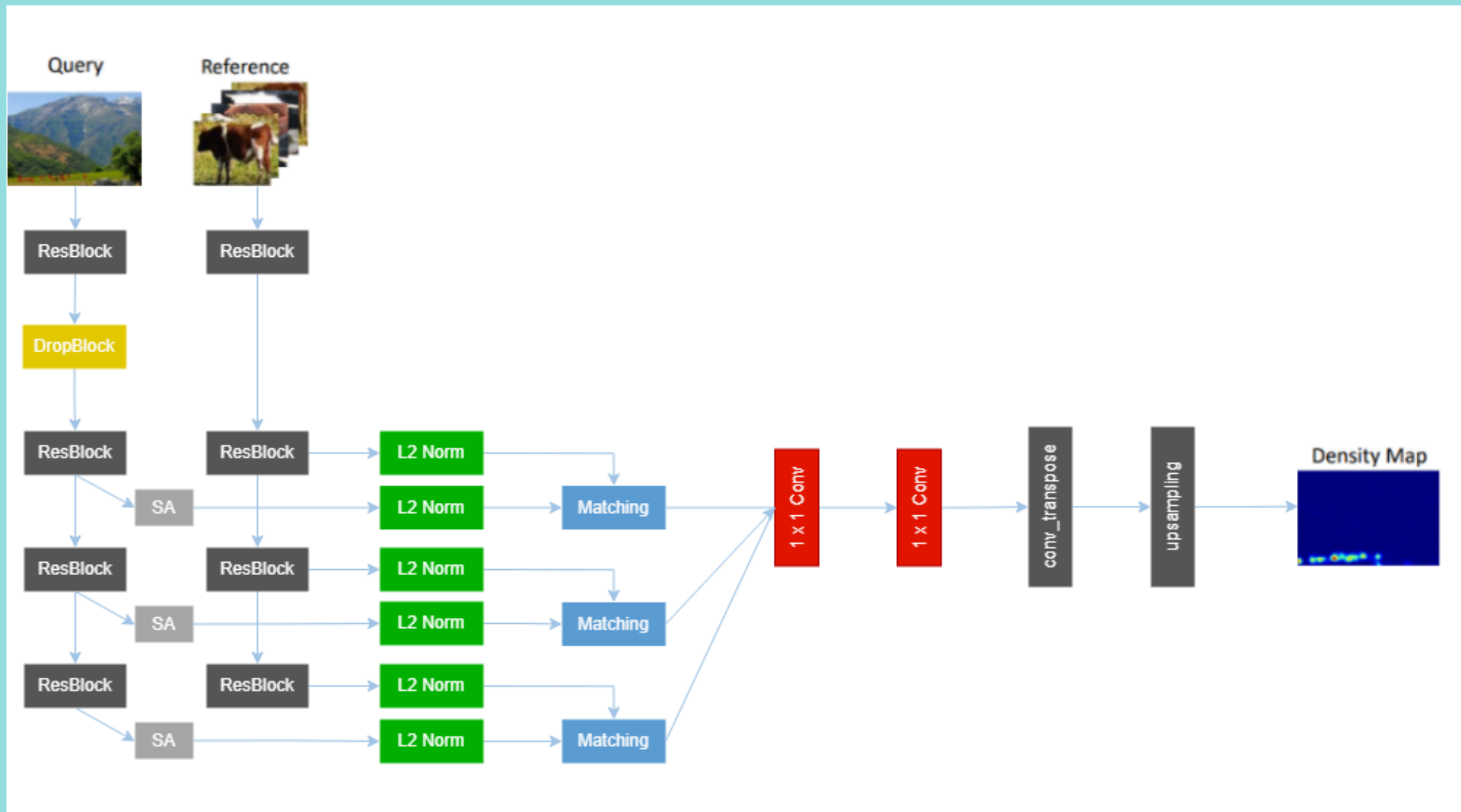
具體方法



改良結果

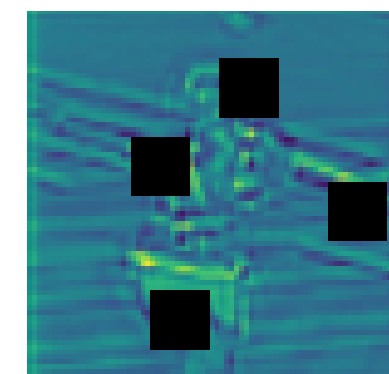
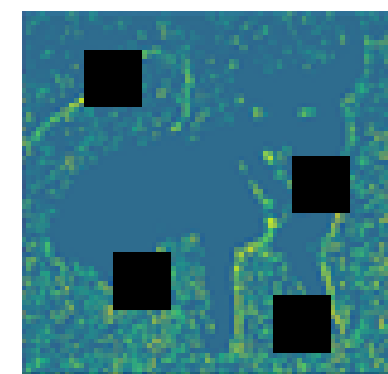
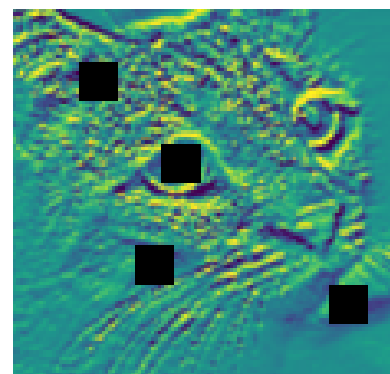
各個位置的 Pixel 可以分開融合不同的 Granularity

改良方法 – DropBlock



改良方法 – DropBlock

具體方法

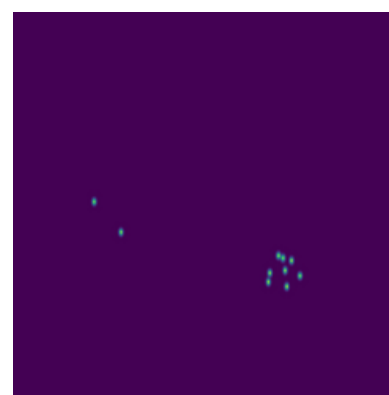


改良結果

增加模型泛化能力，克服 Overfitting

改良方法 – Focal / Huber Loss

具體方法



改良結果

生成的 Density Map 更加聚焦

實驗指標

$$\text{MAE} = \frac{1}{L} \sum_{i=1}^L |\tilde{c}_l - c_l|$$

$$\text{RMSE} = \sqrt{\frac{1}{L} \sum_{i=1}^L (\tilde{c}_l - c_l)^2}$$

$$\text{NAE} = \frac{1}{L} \sum_{l=1}^L \frac{|\tilde{c}_l - c_l|}{c_l}$$

$$\text{SRE} = \sqrt{\frac{1}{L} \sum_{i=1}^L \frac{(\tilde{c}_l - c_l)^2}{c_l}}$$

實作細節

資料集	360 Dataset + Pandora Dataset
Data Augmentation	Mosaic, Flipping, Rotation, Color Jitter, Gaussian Blur, Motion Blur
Batch Size	8
Epoch	120
Optimizer	Adam
Reference Image 數量	7
Learning Rate	5e-5
Focal Weight	15
DropBlock 機率	0.15

實驗結果

	NAE	SRE	Inference Time (FPS)
Baseline	0.842	8.300	24.98
+ DropBlock	0.386	4.835	24.42
+ Deform Feature	0.356	3.909	21.39

註：Baseline = CFOCnet + L2 Norm + Focal Loss + Huber Loss

實驗結果比較

Model	NAE	SRE
FamNet	0.44	6.45
BMNet+	0.27	6.2
CounTR	0.23	7.44
SAFECount	0.25	6.37
SPDCN	0.22	2.19
LOCA	0.19	6.17
MGCAC	0.16	5.68
360VideoCAC	0.35	3.90

實驗結果 - Density Map

Stage 1

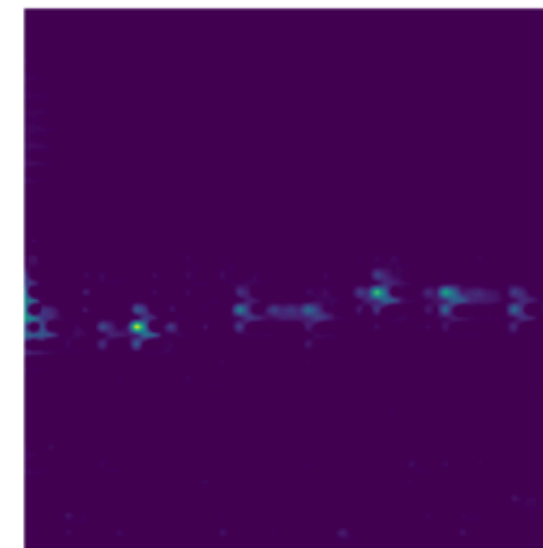
Query Image



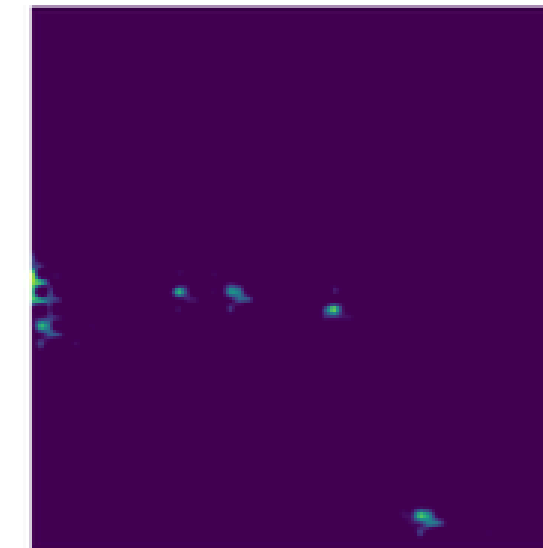
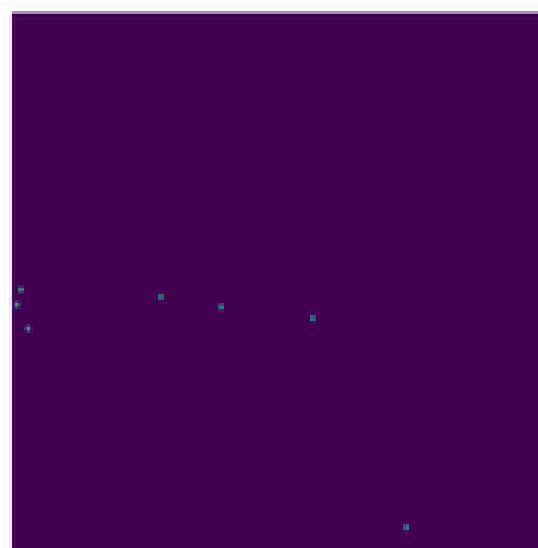
Ground Truth



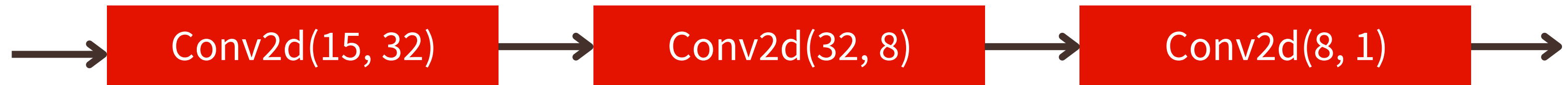
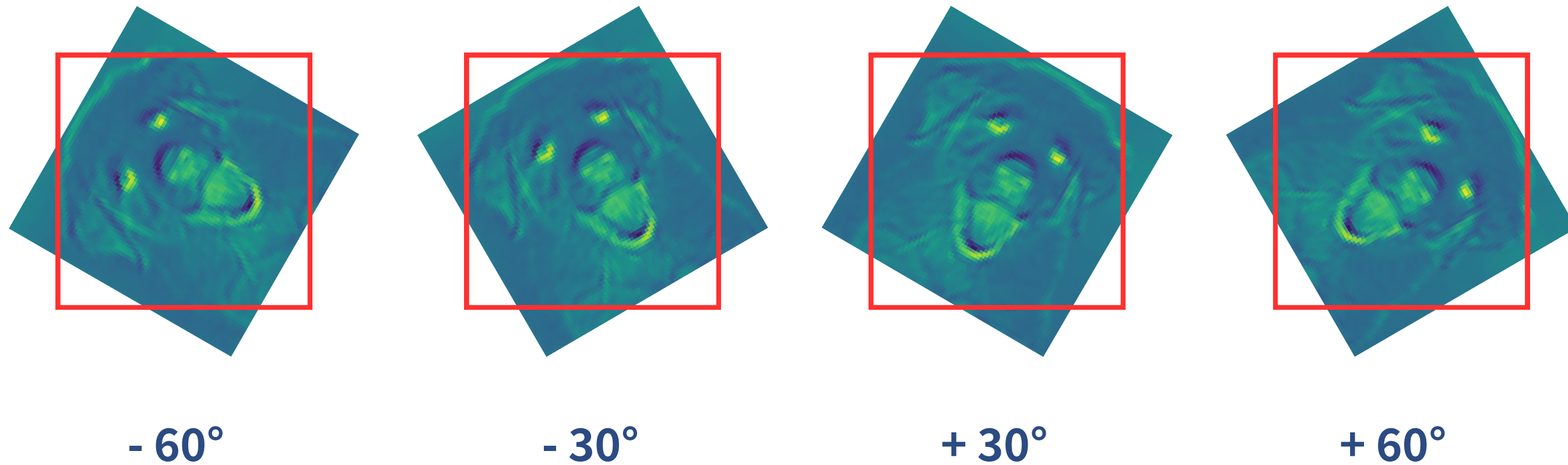
Output



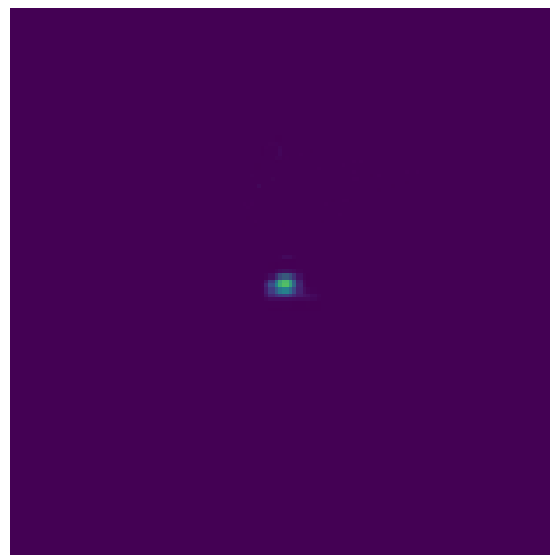
Stage 2



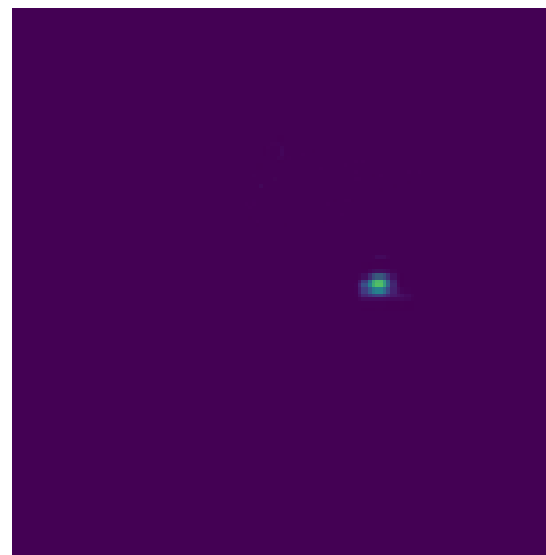
未來規劃 - Rotation Feature



未來規劃 - General Loss



Ground Truth



Output (A)



Output (B)

Huber Loss	(A) 與 (B) 的 Loss 相同
General Loss	(B) Loss > (A) Loss